FinInteract: The Single-Gold Illusion and the Elicitation–Integration Gap in Ambiguous Question Answering

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Abstract

Large language model agents are increasingly deployed for financial information retrieval and hence downstream analytics or investment decisions. But financial terminology is specific, and a question without highly specification may admit multiple defensible readings with materially different values, e.g. Meta Platforms' "operating income" is \$46.75B consolidated or \$62.87B for the Family of Apps segment, but each reading's ground truth is exact and programmatically verifiable. A capable agent should recognize the ambiguity and ask a clarification question instead of committing to a plausible yet unintended reading. However, existing financial benchmarks generally attach only one gold answer per question, failing to distinguish between agents which resolve the ambiguity from those making ungrounded guesses, which we call blind spot *single-gold illusion*. To address this illusion, we release FinInteract, a bilingual (English/Chinese) benchmark of 173 instances that pairs each question with a *default* and an *intended* interpretation (fixed by a disambiguating context) across a five-category ambiguity taxonomy, and grades whether an agent elicits and then integrates the right clarification. Across a range of proprietary and open models, we observe that re-grading the *same* outputs against the intended answer leads to accuracy drop by more than 3 times, verifying the single-gold illusion. Our fine-grained analysis shows that this failure is interaction as models actually answer with over 90% accuracy once the interpretation is resolved but only about 45% under their own clarification, suggesting that the key bottleneck lies on correct clarification and downstream integration. We further include skill metric that localizes failure modes of different financial ambiguities across two well-powered categories (entity scope and metric definition) and three exploratory ones.

1 Introduction

Large language model (LLM) agents are increasingly utilized in financial analysis to answer questions in regulatory filings (Islam et al. 2023; Chen et al. 2021; Reddy et al. 2024), carry out multi-step research and retrieval over corporate disclosures (Bigeard et al. 2025; Choi et al. 2025; Krumdick et al. 2024; Xie et al. 2024). The agent reads a user's question, searches through relevant filings and returns the answer for downstream actions.

However, financial questions are often deceptively underspecified. Financial terminology is highly specific, and a natural language question may map to several defensible readings for different numbers, e.g. in the question *"What was Meta Platforms' operating income?"* (Figure 1), an agent must decide whether the user means the consolidated GAAP figure (\$46.75B for FY2023) or the Family of Apps segment (\$62.87B), both being programmatically verifiable against SEC facts. A capable agent should recognize the ambiguity and follow up with the user instead of directly committing into a plausible but unintended reading.

Existing financial benchmarks cannot measure this as they grade only the final answer by assuming the question is disambiguated (Islam et al. 2023; Chen et al. 2021; Reddy et al. 2024; Krumdick et al. 2024; Xie et al. 2024). Without grounding check, they cannot distinguish whether the agent actually resolves the ambiguity or guesses the common reading. We call such blind spot *single-gold illusion* (Table 13), which is currently underexplored in the financial domain relative to general ambiguous QA literature (Min et al. 2020; Li et al. 2025; Kuhn, Gal, and Farquhar 2023). We study financial ambiguity over three research questions:

- **RQ1**: Do frontier models recognize financial report ambiguity?
- **RQ2**: What financial ambiguity leads to most failures?
- **RQ3**: Can we incorporate interventions at inference time to improve resolution?

Answering these questions requires three design properties. First, the benchmark must draw representative ambiguity from real filings. Second, it must expose fine-grained dimensions of ambiguity through paired default and intended readings. Third, benchmark construction should have low cost and is safe from potential leakage to keep answer out of disambiguating context, with verifiable ground truth validated by domain experts. (Sec. 2).

We release FinInteract, a bilingual benchmark in English and Chinese containing 173 instances, classified into five financial ambiguity categories. Each question Q is paired with a *default* interpretation recording a possible non-expert assumption and an *intended* interpretation fixed by the disambiguating context. By grading the same model outputs

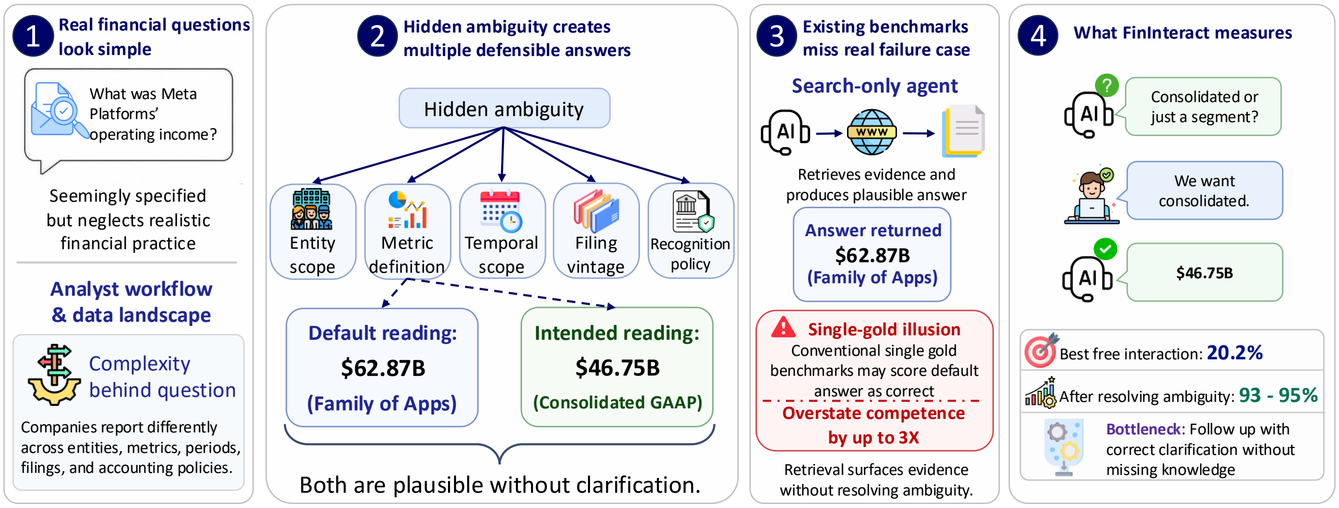

Figure 1: **Evaluating financial agents on ambiguous queries. (1–2)** A real query ("What was Meta Platforms' operating income?") looks fully specified but hides ambiguity across five categories (entity, metric, period, filing, policy), here yielding two defensible answers: a default \$62.87B (Family of Apps segment) and the intended \$46.75B (consolidated GAAP). **(3)** A search-only agent returns the plausible default, which a single-gold benchmark scores as correct, the *single-gold illusion* that overstates competence by up to $3\times$. **(4)** FinInteract instead grades whether the agent *asks*: the GPT-5 agent reaches 20.2% under free interaction versus 93–95% once the ambiguity is resolved, so the gap lies in interaction, not missing knowledge.

against the default and intended interpretations, we effectively expose this single-gold illusion. FinInteract includes skill metric to localize which clarification ability the model lacks, and evaluates the agents under a ReAct protocol where the user simulator answers only from the disambiguating context (Figure 2) to measure the agent's quality of clarify-then-integrate action.

Our evaluation answers the research questions in turn. First, while frontier models detect ambiguity, they rarely resolve it effectively, as they ask the correct clarification question but still **failing to convert into the correct answer at the end** (Sec. 4.4). Second, **policy-related ambiguities** are the least resolved, as none of the proprietary models would initiate clarifications when facing this sparse category. Third, **inference intervention facilitates ambiguity resolution**, increasing accuracy from about 44% to above 90%, suggesting that this benchmark is not only diagnostic but also actionable. (Appendix C). This paper makes three contributions

1. **Problem definition.** We formalize ambiguity in financial questions with five categories grounded in standard accounting and disclosure practice (Sec. 2).

2. **Benchmark.** We introduce FinInteract, a bilingual financial ambiguity dataset containing 173 instances, each paired with a default and an intended interpretation verified by domain experts. (Sec. 3).

3. **Empirical findings.** Our evaluation shows that existing models commonly fail to return correct final answer even when they have correctly clarified the intention, which can be improved via intervention at inference time. (Sec. 4).

2 Problem Formulation

Given an ambiguous financial question instance (example in Table 1), it should contain the ambiguous question over a company filing Q, the disambiguating context C which is the minimal set of scope or definitional constraints that collapses Q into the unique intended answer A under intended interpretation $\mathcal{I}_{\text{int}}$. The ambiguous question typically contains correct answer $A_d \neq A$ under the default interpretation $\mathcal{I}_{\text{def}}$ which a non-expert would assume when reading Q. Specifically, $\mathcal{I}_{\text{int}}, \mathcal{I}_{\text{def}}$ are structured records specifying the financial ambiguity taxonomy defined below.

Table 1: This example of ambiguous financial question contains three plausible entity scopes, i.e. consolidated total and two reportable segments, leading to an entropy of 1.58 bits. Without the disambiguating context C, a model's default interpretation leads to the wrong answer.

Field	Value
Q	What was Meta Platforms' operating income?
C	Total company (consolidated) results under U.S. GAAP for the year ended December 31, 2023.
A (intended)	\$46.75B *[consolidated GAAP]*
A_d (default)	\$62.87B *[Family of Apps segment]*
Ambiguity category	Entity scope
H_0	1.58 bits
Intended span	"Income from operations for 2023 was \$46.75 billion. . . "
Default span	"Family of Apps. . . Income from operations \$62,871. . . "

Financial ambiguity taxonomy. We derive five categories of financial ambiguity from standard accounting and disclosure practice (Table 2). Each instance should normally exercise one primary category with one or two optional secondary categories in case of additional ambiguity.

Table 2: Five-category financial ambiguity taxonomy.

Category	Definition
Temporal scope	Ambiguity over which time period is intended: fiscal year ending September vs. calendar year; Q3 results vs. full year; TTM vs. point-in-time.
Metric definition	Ambiguity over accounting basis: GAAP net income vs. non-GAAP adjusted income; organic revenue vs. as-reported; operating EBITDA vs. net income.
Entity scope	Ambiguity over consolidation level: segment vs. consolidated total; parent-attributable vs. full consolidated; Class A vs. Class B shares; A-shares vs. H-shares.
Filing vintage	Ambiguity over filing version: original 10-K vs. amended 10-K/A; as-reported vs. restated; preliminary earnings release vs. final filing.
Recognition policy	Ambiguity over accounting policy: revenue recognized point-in-time vs. over time; gross vs. net revenue; capitalized vs. expensed R&D.

Disambiguation entropy. We quantify how ambiguous an instance is using the prior confidence entropy $H_0 = \log_2 k$, where k denotes the number of plausible interpretations of Q without clarifying interaction, e.g. the example in Table 1 has three plausible interpretations, hence contributing to an entropy of $H_0 = \log_2 3 \approx 1.58$ bits. This further serves as our metric to evaluate difficulty and efficiency in Sec. 3.5. While both A and A_d are defensible from publicly available filings, an agent should be able to identify the ambiguity by reading Q alone without C, where this pairing between default and intended interpretation generalizes the target-distractor design from InteractComp (Deng et al. 2026) to the financial domain.

Agent Interaction Model. We follow InteractComp's ReAct framework (Yao et al. 2023) where the agent may issue three types of actions, namely **search**(Q) to retrieve a passage from the filing corpus, **interact**(Q) to present a yes-or-no question to the simulated user who answers based on C, and **answer**(Q) to submit a final answer. Specifically, we evaluate agents in three primary modes (Table 3) and four ablation modes. We additionally include a category-conditioned interaction mode, described in Appendix C.7.

3 Methodology

FinInteract extends formulation in Sec. 2 as a semi-automated ambiguous financial QA corpus built from public filings. Each carries a programmatically verifiable answer, a paired default and intended interpretation produced by an LLM construction pipeline with minimal human annotation and validated by domain experts. Figure 2 shows a worked instance and how the benchmark scores an agent's two resolution paths.

Table 3: We evaluate interaction modes including answer-only (A), answer with search (A+S) and answer with search and interaction (A+S+I).

Mode	Description
A	Pure parametric recall
A+S	Oracle retrieval augmented
A+S+I	Standard full interaction
Always-ask	≥ 1 follow-up question
Category-oracle	Provide primary category
Template-oracle	Fixed human-written question
Enumerate	Lists all interpretations and answers

3.1 Design Goals and Principles

Design goals. FinInteract targets four goals: **(G1) Representative coverage** to draw real ambiguity from public filings. **(G2) Fine-grained diagnosis** to contain diverse possible readings so that evaluation can localize which clarification ability a model lacks. **(G3) Low-cost, leak-safe construction** to build and verify the corpus at scale in a semi-automated setting. **(G4) High, human-validated quality** to validate the instances and judges by domain experts with minimal annotation cost.

Design principles. To satisfy the four goals, we build FinInteract under four principles: **1): Easy to verify, Ambiguous to resolve** to create programmatically verifiable ground truth using SEC XBRL facts and CNINFO-structured data but the question cannot be correctly resolved without disambiguating context C, **2): No yes/no questions** so that the agent has to deliver a grounded reasoning support, **3): Includes company name** to ensure no ambiguity in entity, i.e. capitalized proper noun for English and CJK company name for Chinese, and **4): Answer is not in disambiguating context** C so the agent cannot take shortcut by copying directly from the context.

3.2 Data Sources

We build the instances from official filings. For English instances which take about 80% of the passage pool, we build them from SEC filings, DocFinQA long-context 10-K/10-Q passages with verified QA pairs and XBRL-derived facts for S&P 500 and Russell 1000 companies within the financial year of 2024 to 2025. For Chinese instances, we build them from A-share annual reports of SSE 50 and CSI 3000 constituents using CNINFO and akshare structured APIS. We avoid reusing published QA pairs directly to reduce memorization risk, though we do not claim absolute contamination safety. The current evaluation corpus contains 173 instances over 61 companies within 2022 to 2025 filings, labelled with ambiguous taxonomy from Sec. 2, with detail breakdown in Table 4. Specific details regarding passage counts and difficulty distribution are included in Appendix H.

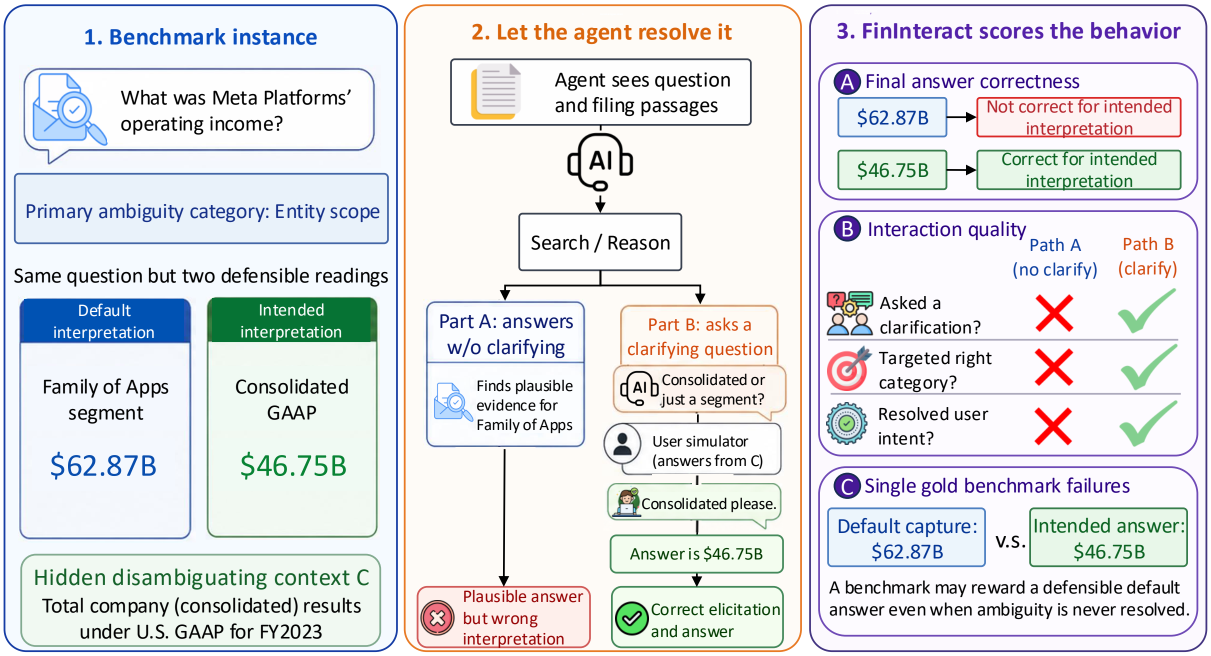

Figure 2: **FinInteract rewards actions that resolve the intended interpretation.** **(1)** Each instance carries a primary ambiguity category and two defensible readings, a default (Family of Apps, \$62.87B) and an intended (consolidated GAAP, \$46.75B), separated by a hidden disambiguating context C. **(2)** The agent either answers without clarifying (Part A, a plausible but wrong reading) or asks a question that a user simulator answers only from C (Part B, correct elicitation). **(3)** FinInteract scores three signals: final-answer correctness against the intended interpretation, interaction quality (did it ask, target the right category, and resolve intent), and the single-gold failure of rewarding the default.

Table 4: FinInteract dataset statistics. Disambiguation entropy $H_0 = \log_2 k$ over k plausible interpretations.

Property	Value	Property	Value
Total instances	173	**Category**: entity	94
Language: EN	53	metric	63
ZH	120	recognition	9
Source: EDGAR	50	temporal	7
CSRC-CNINFO	120	filing	0
DocFinQA	3	**# cat.**: single	128
Filing: 10-K	53	two	45
annual report	120	H_0: mean	2.20
Filing yr: 2022	40	median	2.00
2023	53	range	1.0–3.6
2024	58	**Blind-solve rate**	0.7%
2025	19	**Unique companies**	61

3.3 Curation

We curate each instance from a three-role pipeline, where a **constructor** agent using GPT-5 generates the full (Q, C, A) triple together with the default and intended interpretations and category label from a filing passage. Specifically, we include category specific instructions to prevent drifting towards categories that are easy to generate. Next, we conduct a **sanity check** on code level to reject malformed passages instantly without API calls. Lastly, we employ an **adversarial verifier** to run multiple trial rounds to answer the question Q without disambiguating context C and reject instance where more than two trials produce the intended answer and interpretation simultaneously. Details regarding the prompt, sanity checking codes, output schema and cost are included in Table 15, Appendix H and Appendix G.

3.4 Validation

We employ finance-literate annotators, including Chartered Financial Analyst (CFA) charterholders and licensees from Hong Kong Securities and Futures Commission (SFC), to review instances to check whether 1) question Q is ambiguous and 2) disambiguating context C uniquely identifies the intended reading without stating the answer and 3) both intended and default answers are correct. We record a substantial agreement with an average 0.85 Gwet's AC1 (Gwet 2008), indicating strong chance-corrected consensus that, unlike Cohen's κ, stays reliable under the heavy skew toward "yes" labels. The full protocol and annotation details are in Appendix H.

3.5 Metrics

We include standard metrics from ambiguous QA and interactive retrieval literature (Chen et al. 2021; Deng et al. 2026; Islam et al. 2023; Wei et al. 2025) to directly compare FinInteract to prior benchmarks.

Accuracy (%) measures the fraction of instances whose final answer matches the *intended* interpretation, graded by

GPT-4o-mini with financial domain tolerance including measurement difference, entity equivalence across different stock markets, current normalization or fiscal year matching.

Default-capture rate (%) is the fraction whose final answer instead matches the *default* interpretation, which measures how often a model returns the naive reading so we can reconstruct what a conventional single-gold benchmark would have scored (Sec. 4.3).

Interaction diagnostics includes the interaction rate (IR) which measures proportion of instances with more than one clarification question, average turns of clarification (Round), and accuracy on predicting the true ambiguity category (AC@1). Specifically, we report AC@1 by whether the question names any true category and whether it can name the single most important category.

Confidence measures the Expected Calibration Error (ECE) between model confidence and correctness to capture agent's overconfidence when reading the ambiguous queries.

Full metric definitions, including the accuracy grading tolerance, the AC@1 decomposition, and a supplementary clarification-efficiency score (DisE$^+$), are in Appendix A.

4 Experiments

4.1 Evaluation Setup

Agent architecture. We evaluate agents under the ReAct (Yao et al. 2023) framework under seven configurations from Table 3. The **interaction** action presents a yes-or-no question to the simulated user where the user has to respond "yes", "no" or "I don't know".

Prompting strategy. FinInteract evaluates interactive agents under different interaction protocol, including answer-only, answer with search, answer with search and interact, and the category conditioned policies. All models run under the same ReAct system prompt. Additional information is in Appendix C.7.

Models. We evaluate over a diverse models such as proprietary models including GPT family with GPT-5.6-sol, 5, 4o and 5-mini, Claude-Somnet-5, Gemini-3.5-flash, Grok-4.5, DeepSeek-V4-flash, GLM-5.2 and Kimi-K3, and open source models including Qwen3 family including 4B, 8B, 14B, 32B and two mixture-of-experts models namely Qwen3-30B-A3B and Qwen3.5-35B-A3B. We evaluate the proprietary model via OpenRouter.

4.2 Overall Performance

Table 6 shows that even frontier models are vulnerable to financial report ambiguity. Specifically, we see that no models exceeds 44% accuracy on the intended interpretation under free interaction mode, consistent across proprietary and open source models. The opportunity to interact does not help to improve accuracy, as the interaction difference Δ is non-positive for most models, indicating that existing models do not fail in asking correctly, but in converting the clarified intention into the correct answer. This is supported by the 95% interaction rates and high AC@1 indicating that the models do recognize what kind of ambiguity it is to ask the correct clarification question. Specifically, we conduct further ablation studies in the interaction by forcing multiple interaction

Table 5: Forcing more clarification does not help with accuracy, but *resolving* the ambiguity lifts every model to 93-95%. This gap suggests that it is the model's inability to elicit Cv itself but not missing knowledge or grounding (Same abbreviation as Table 3).

Model	A	A+I	Forced(4)	**Oracle**
GPT-5	0.6	20.2	1.5	**95.4**
GPT-4o	0.6	4.6	0.0	**93.1**
GPT-5-mini	0.6	0.0	0.6	**95.4**
Qwen3-30B	1.2	11.6	–	**90.2**
Qwen3.5-35B	0.6	28.9	–	**91.9**

rounds and directly providing the disambiguating context C as oracle, Table 5 shows that forcing more rounds of clarification actually lowers the model accuracy, which may imply agents getting lost in multi-turn conversations (Laban et al. 2026). Moreover, under the oracle scenario, the models can answer at about 93 - 95% accuracy consistently. As the context C does not contain final answer A, the improvement suggests that the gap is not missing knowledge, but the ability to disambiguate appropriately for downstream reasoning. We include the full finding in Appendix B. **Finding 1: Models are capable of observing the ambiguity and answering correctly in oracle-clear intention, but fails to guide itself during intermediate conversion between updated intention and correct answer.**

4.3 Failure of Existing Benchmarks

By re-grading models on default readings from existing benchmark and the intended reading FinInteract curates and verifies, we observe that the accuracy differs significantly (Table 6) with a consistent inflation on the default scenario. This suggests that if existing benchmarks are curated without specific linguistic updates, model performance in financial QA might be conflated. We include details of the findings along with human-annotated study in Appendix B. **Finding 2: Conventional single-gold financial QA benchmarks fail because building benchmarks with only one gold answer would likely encode default reading but not human-preferred intentions.**

4.4 Systematic failure from recognition policy

We extend the study into each ambiguity group. Table 6 shows that in the high AC@1 shown in Table 6, while models are capable of determining ambiguity category in most cases, it consistently fails in the recognition policy group. This motivates us to study this blind spot of existing models by examining the following case study. Given the question "What were General Electric's revenues?", the answer is either \$67.95B or \$26.79B under different recognition policies, the former being GAAP revenue that combines over-time service recognition with point-in-time product recognition, and the latter counting only revenue recognized at a point in time. Models fail here as they consistently return a single figure without asking for the expected recognition basis, defaulting to a generic assumption that "revenue" is well defined

Table 6: **(a)** Main results, reporting accuracy (%) against the *intended* interpretation per mode, with Δ the +Interact minus +Search difference (full-scale rows are N=173 and the cross-vendor frontier rows, IR and ECE blank, are a preliminary N=50 pilot). **(b)** Re-grading the same outputs against the intended versus the default reading a conventional benchmark would treat as gold (Same abbreviation as Table 3). **(c)** Per-category targeting (AC@1), where recognition policy is a shared blind spot (= 0 for every model). Category sizes: entity (Ent.) 94, metric (Met.) 63, recognition (Rec.) 9, temporal (Tmp.) 7.

(a) Main results

	Acc (%)		Answer+Search+Interact				
Model	A	A+S	A+S	Δ	IR	AC@1	ECE
Proprietary models							
GPT-5.6-sol	10.0	36.0	30.0	-6	–	.98	–
GPT-5	0.6	6.4	**20.2**	+13.8	98.8	.86	.53
GPT-4o	0.6	12.1	4.6	-7.5	99.4	.94	.82
GPT-5-mini	0.6	0.0	0.0	0.0	96.0	.94	.00
Claude-Sonnet-5	4.0	**46.0**	**44.0**	-2	–	.98	–
Grok-4.5	2.0	40.0	32.0	-8	–	.87	–
GLM-5.2	12.0	40.0	40.0	0	–	.96	–
Kimi-K3	16.0	38.0	34.0	-4	–	.98	–
DeepSeek-V4-flash	12.0	28.0	24.0	-4	–	.96	–
Gemini-3.5-flash	14.0	6.0	30.0	+24	–	.96	–
Open source models							
Qwen3-4B	0.0	8.1	12.1	+4.0	28	.85	–
Qwen3-8B	0.6	14.4	24.9	+10.5	15	.88	–
Qwen3-14B	0.0	18.5	22.5	+4.0	36	.74	–
Qwen3-32B	1.2	27.2	8.4	-18.8	61	.96	–
Qwen3-30B	1.2	28.3	11.6	-16.7	72	.90	–
Qwen3.5-35B	0.6	**35.8**	28.9	-6.9	36	.81	–
Human baseline (N=50)							
Human	–	–	**30.0**	–	100	.72	–

(b) Single-gold illusion

Model	Mode	Intended (true Acc)	Default (single-gold)	Δ
GPT-5	A	0.6	2.3	+1.7
	A+S	6.4	10.4	+4.0
	A+S+I	**20.2**	12.1	−8.1
GPT-4o	A	0.6	0.6	0.0
	A+S	12.1	**37.0**	**+24.9**
	A+S+I	4.6	6.4	+1.8
GPT-5-mini	A	0.6	0.0	−0.6
	A+S	0.0	0.0	0.0
	A+S+I	0.0	0.0	0.0

(c) Per-category targeting (AC@1)

Model	Ent.	Met.	Rec.	Tmp.
GPT-5	.89	.92	**.00**	1.00
GPT-4o	1.00	.97	**.00**	1.00
GPT-5-m	.99	1.00	**.00**	1.00
Q3-30B	.93	.96	**.00**	1.00
Q3.5-35B	.85	.87	**.00**	1.00

and ignoring the ASC 606 practice of recognizing revenue at a point in time versus over time. This suggests that the domain-specific practice might not align too well with existing model's generalized behavior, hence suggesting the need of specific tuning. We attach details and further case studies in in Appendix B. **Finding 3: Agents are not well-trained to notice ambiguity on financial-specific policies.**

4.5 Ambiguity category as training signal

While observing that the diagnosis is descriptive, we ask whether we can leverage the ambiguity category as training signal. We conduct conditioned training on the model, including fine-tuning on different categories followed by reinforcement learning on a Qwen3-4B policy, teaching hte model to interact and target the correct categoy. Within the held-out split, we seek that the signals already jump significantly at supervised stage (Table 7, Figure 3a), and a full multi-turn reinforcement training further confirms that the episode is trainable end to end, with the clarifying turns rising from 5.7 to 8.6 (Figure 3b). The gain concentrates on the data-abundant categories, so the signal is actionable but still blinded in recognition policy in Sec. 4.4. Additional details are included in Appendix C.7. **Finding 4: FinInteract is a trainable signal to address financial ambiguity.**

Table 7: RLVR ladder on FinInteract using Qwen3-4B with held-out n=51 showing that we can train policy to clarify ambiguity using leak-proof AC@1 and interaction.

Stage	AC@1	Interaction	Accuracy	Reward
Base	25.0	54.9	84.3	0.96
+ SFT	68.6	**100**	88.2	1.28
+ KTO	**70.6**	**100**	84.3	1.27
+ GRPO	66.7	**100**	88.2	**1.30**

5 Discussion

In our study, we observe that **categories are different learning problems.** While some clarification abilities are solved by scale, others, notably recognition basis, resist it, so a single aggregate score hides where the difficulty sits and curricula should budget data per category. On the other hand, **elicitation is a trainable skill current models lack.** Progress will come less from larger backbones, which barely move interaction accuracy, than from policies and objectives that reward asking on the right category at the right time. Nevertheless, even by including annotation for domain experts, it is still possible to retain **ambiguity in Human-AI collaboration** because models confidently return the default reading, motivating a safe behavior of asking instead of answering, hence the development of metrics that credit well-timed clarification and calibration process. While the current FinInteract

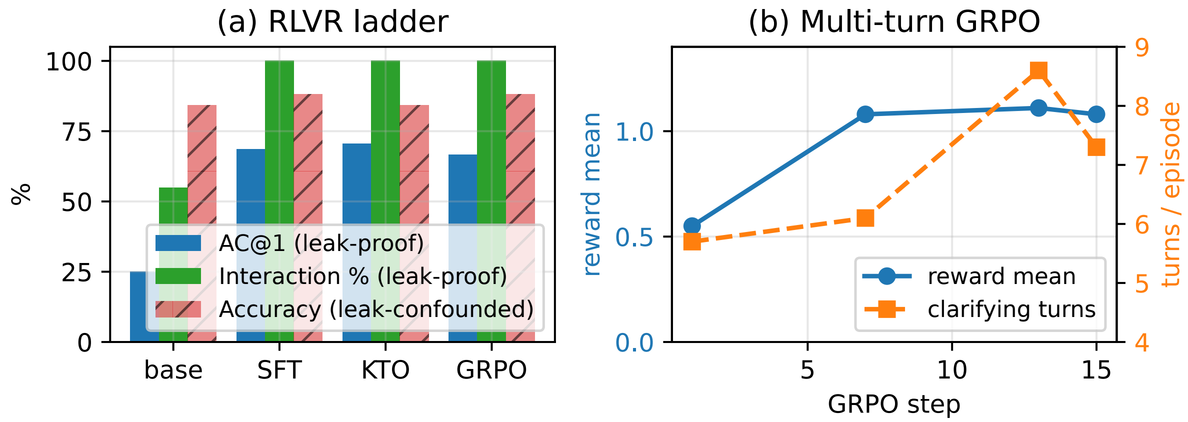

Figure 3: Training on FinInteract. (a) Using AC@1 and interaction as training signal already improves performance in SFT. (b) True multi-turn GRPO on the 4B policy: mean reward and clarifying turns rise together over training.

is semi-automated, ideally a benchmark should be able to extend itself on categories and paired default/intended construction whose evidence is abundant, so as to **transfer to further categories, languages and other high-stakes domains** where ambiguity is commonly observed.

We note the following limitations of this study: **1) Scale** where a 173 instances corpus is considerably small given the need of construction rigor, **2) Language balance** where current corpus split between English and Chinese is 53 : 120, which could benefit models pre-trained in certain language, **3) Dominating categories** with entity scope (n=94) and metric definition (n=63), compared to the blind spot recognition policy with only n=9, which may suggest existing model is not well-trained enough to even generate such instance in the first place, **4) Generator and Evaluator bias** due to API budget constraints, as we hire GPT-5 family mainly for construction and simulation, which may bias towards the performance of GPT-5 family. **5) Data contamination** whereas instances are freshly constructed from FY2022-2025 facts where recently released models might have been pre-trained on. We include extended discussion in Appendix F.

6 Related Work

We include the most relevant works to FinInteract and defer additional financial and general-domain benchmarks as well as more widespread related work in Appendix D.

Financial QA benchmarks such as FinanceBench (Islam et al. 2023), FinQA (Chen et al. 2021), and DocFinQA (Reddy et al. 2024), together with recent agentic successors, enforce single-gold, single-turn evaluation and test numerical reasoning or retrieval without exploring whether a query is under-specified. To our knowledge, FinInteract is the first to make query ambiguity the central focus through paired default/intended interpretations and multi-turn interaction in the financial domain. We include comparison with existing financial QA benchmarks in Appendix D.

Ambiguous QA and clarifying questions. In the general domain, AmbigQA (Min et al. 2020), CLAMBER (Zhang et al. 2024), and QuestBench (Li, Kim, and Wang 2026) study whether a model identifies under-specification and asks the minimal clarifying question, building on a long clarifying-question line in retrieval and dialogue. InteractComp (Deng et al. 2026) is our direct methodological ancestor, which we extend to finance with a domain-specific ambiguity taxonomy, per-category targeting (AC@1), and programmatically verifiable XBRL-grounded evidence pairs.

Multi-turn interaction and selective answering. Large models often *lose track* over multi-turn conversations (Laban et al. 2026), and calibrated systems frequently do better by answering *selectively* or abstaining under ambiguity (Cole et al. 2023; Kuhn, Gal, and Farquhar 2023). Our findings echo this in the financial setting, where forcing extra clarification degrades accuracy, but FinInteract turns the observation into a controlled, per-category diagnosis of *which* clarification an agent should issue and *whether* issuing it resolves the query, instead of only measuring whether more turns help.

7 Conclusion

We introduced FinInteract, the first bilingual benchmark for evaluating financial search agents on genuinely ambiguous queries. The five-category financial ambiguity taxonomy, paired default/intended interpretation design, and construction pipeline with adversarial verifier yield 173 high-quality instances to resist trivial disambiguation. Our central finding is that the key bottleneck of existing agent's ability to handle financial QA ambiguity is interaction capability, but not financial knowledge. This capability contains two parts, i.e. the ability to ask for the right clarification and to update internal belief of the original answer. Specifically, when the ambiguity is fully solved, agents can answer with far higher accuracy at 93-95% accuracy. This amplifies the stagnation identified in InteractComp in the general domain, where the more terminology- and domain-specific financial industry requires specialized benchmark. We further turn this diagnosis into a training intervention in Sec. 4.5 to showcase that FinInteract also serves as an informative training signal, and release FinInteract with full construction, evaluation, and training code for reproducible follow-on work.

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A Metric Details

This appendix expands the metrics defined in §3.5.

Accuracy grading. Accuracy is the primary ranking metric, mirroring InteractComp, BrowseComp, and FinanceBench. A GPT-4o-mini grader marks a final answer correct against the intended interpretation under financial-domain tolerance: a $\pm 1\%$ numeric tolerance, entity/ticker equivalence, and currency normalization, with a fiscal-year mismatch counted as wrong.

Interaction targeting (AC@1). For each interact action, a GPT-4o-mini classifier decides whether the clarifying question targets a true ambiguity category for that instance. We report the targeting accuracy AC@1 alongside the any-category, wrong-category, and generic-ask rates. Because a single clarification is often compound (one question can name an entity, a period, and a metric at once), AC@1 has two forms: a lenient *touch* form (the question names any true category) and a strict *central* form (the single most-central category is a true category). The touch form is validated against human annotators (§3.4). These diagnostics characterize *how* a model interacts and do not rank models.

Calibration. The Expected Calibration Error (ECE) between stated confidence and correctness uses five equal-width confidence bins.

Clarification efficiency (DisE$^+$). As a single supplementary number folding accuracy and clarification cost together, we report the efficiency of *successful* clarification,

$$\mathrm{DisE}^+ = \mathbf{1}[\text{correct}] \times \mathbf{1}[n_{\text{asks}}{>}0] \times \frac{H_0}{n_{\text{asks}}},$$

which is positive only when the agent asked at least one question and then answered correctly, and decreases with redundant asks. We restrict it to the asked-and-correct case to avoid the perverse incentive of an all-instances variant, which would award its maximum to a confident *zero-ask* answer, rewarding the very behavior the benchmark penalizes. DisE$^+$ is a diagnostic of clarification efficiency, never the ranking metric.

B Detailed Results for the Core Findings

This appendix gives the full tables, figures, and findings summarized in §4.3, §4.2, and §4.4.

B.1 The Single-Gold Illusion

Before analyzing model behavior, we establish the core motivation empirically: *a conventional single-gold financial-QA benchmark cannot see the failure FinInteract measures.* Existing benchmarks (FinanceBench, FinQA, DocFinQA) attach one gold answer per question. Built from the same filings, such a benchmark would almost always adopt the *default* reading, the headline figure a non-expert assumes, as that gold. Our paired construction lets us quantify the consequence directly: every instance carries both an intended and a default answer, so we re-grade the *identical* model outputs under both conventions (Table 6).

Finding 1: A single-gold convention over-states competence by up to $3\times$, and the gap is the benchmark's reason to exist. In the retrieval-augmented regime where current single-gold benchmarks operate, the larger models return the *default* reading more often than the intended one: GPT-4o is graded 37.0% correct against the default but only 12.1% against the intended answer, a single-gold benchmark would report $3.1\times$ its true competence. The model is not reasoning to the right answer; it is confidently producing a well-grounded answer to the *wrong reading of the question*, which a single-gold benchmark rewards and therefore cannot detect. (GPT-5-mini is the instructive exception: it captures *neither* reading because it asks rather than commits, so single-gold grading does not flatter it, but nor does it answer.) Critically, interaction is the *only* mode where intended accuracy exceeds default-capture (GPT-5: 20.2 vs. 12.1): asking is what converts a default-anchored guess into the correct intended answer. This is precisely the capability single-gold benchmarks are structurally blind to, and what FinInteract is built to measure.

Human validation of the default. The single-gold argument assumes a conventional benchmark builder, seeing only the question and passage, would record the *default* reading as gold. We test this directly: five annotators, shown Q and the passage but not the resolving context C, independently record the answer they would treat as gold. They converge strongly (93% of items unanimous; mean inter-annotator